

L3: Degree Distribution of G(n,p) Model

Here is a well-known fact that you may have learned in a probability course: the Binomial distribution can be approximated by the Poisson distribution as long as p is much smaller than 1. In other words, this approximation is true for sparse networks in which the average degree $p \cdot (n - 1)$ is much lower than the size of the network n . The Poisson distribution is described by:

$$p_k = e^{-\bar{k}} \cdot \frac{\bar{k}^k}{k!}, \quad k = 0, 1, 2, \dots$$

$$\bar{k} = p \cdot (n - 1)$$

$$\sigma_k^2 = \bar{k}$$

Because of the $\frac{1}{k!}$ term, p_k decreases with k faster than exponentially.

You may ask, why to use the Poisson approximation instead of the Binomial($n-1, p$) distribution?

The reason is simple: the Poisson distribution has a single parameter, which is the average node degree $\bar{k}$.

The visualization shows the degree distribution for a network in which the average degree is 50. As we increase the number of nodes n , we need to decrease the connection probability p so that their product remains constant. Note that the Poisson distribution is a rather poor approximation for $n=100$ (*because the average node degree is half of n*) but it is excellent as long as n is larger than 1000.

Food for Thought

Try to derive mathematically the Poisson distribution from the Binomial distribution for the case that p is much smaller than 1 and n is large. If you cannot do it, refer to a textbook or online resource for help.